

Tue-05-13-2025  
Conservation Eq<sup>n</sup>

## Introduction

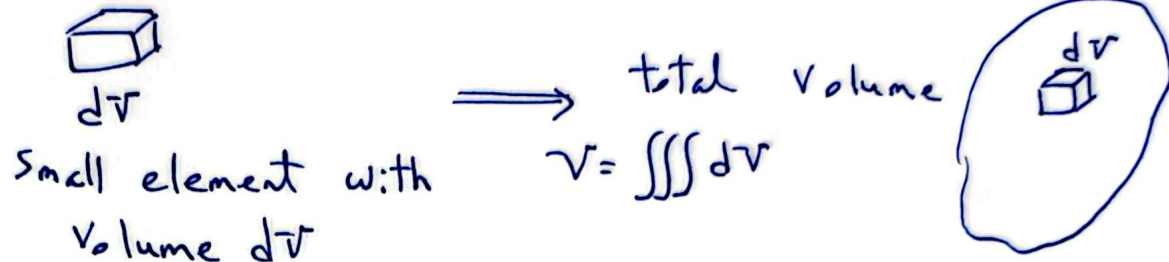
## References & abbreviations used

1. **AE-BC** – Aerodynamics for Engineers, (6<sup>th</sup> edition), John J. Bertin & Russell M. Cummings - course text
2. **FA-JA** – Fundamentals of Aerodynamics, (6<sup>th</sup> edition), John D. Anderson Jr.
3. **FA-KC** – Foundations of Aerodynamics, (5<sup>th</sup> edition), Arnold M. Kuethe & Chuen-Yen Chow
4. **AA-DA** – Applied Aerodynamics: A Digital Text Book - from Desktop Acronautics Inc. <http://docs.desktop.acro/appliedacro/preface/welcome.html>,
5. **IAFV-JGL** – Introduction to Aerospace Flight Vehicles, J. Gordon Leishman
6. **MFMO** – Multimedia Fluid Mechanics Online, General editor G. M. Homsy
7. **AFM-VD** – Album of fluid motion, Milton Van Dyke
8. **VF-JSME** – Visualized Flow, by The Japan Society of Mechanical Engineers
9. **AE-HCCV** – Aerodynamics for Engineering students, E.L. Houghton, P.W. Carpenter, Steven H. Collicott & Daniel T. Valentine

### Other abbreviations used

1. **TEq.** - Equation number in text – Aerodynamics for Engineers, (6<sup>th</sup> edition), John J. Bertin & Russell M. Cummings
2. **TSec** - Section number in text – Aerodynamics for Engineers, (6<sup>th</sup> edition), John J. Bertin & Russell M. Cummings

## Section 1 – Conservation Equations



## Control Volume

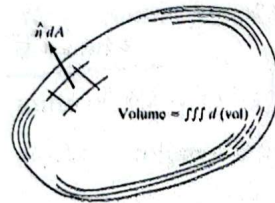


Figure 1.3: Fig. 2.5 from AE-BC

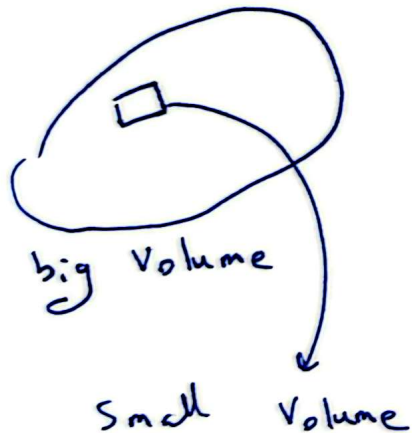
- ▶ Control volume - fixed volume in space through which fluid flows
- ▶  $A$  - surface of the control volume
- ▶  $d(\text{vol})$  - elemental control volume
- ▶  $dA$  - surface of the elemental volume  $d(\text{vol})$
- ▶  $\hat{n}$  - unit normal to  $dA$
- ▶ Therefore, Volume =  $\iiint d(\text{vol})$

Area  $\rightarrow dA$

Every area has two parameters to be represented:

- 1) area magnitude ( $dA$ )
- 2) area normal vector ( $\hat{n}$ )

$$\vec{dA} = dA \hat{n}$$



Area:  $A$

$d(\text{vol})$ : element volume

$dA$ : surface of the element volume

$$\dot{m} = \rho A \vec{V} \quad \text{with } \vec{V} \text{ labeled as } v_{el}$$

$$Q = \frac{\dot{m}}{\rho} = A \vec{V} \quad \text{with } \vec{V} \text{ labeled as } v_{el}$$

$$dQ = \vec{V} dA = \vec{V} \cdot \vec{n} dA$$

### Volumetric flow rate

- ▶  $\frac{dQ}{dA} = \hat{n} dA$  - volume rate of flow through the differential surface  $dA$ . Note,  $\frac{dQ}{dA} = \hat{n} dA$
- ▶ units of  $dQ \rightarrow \text{m}^3 \text{s}^{-1}$
- ▶  $dQ \Delta t$  - volume of fluid flow through a surface  $dA$  in time  $\Delta t$

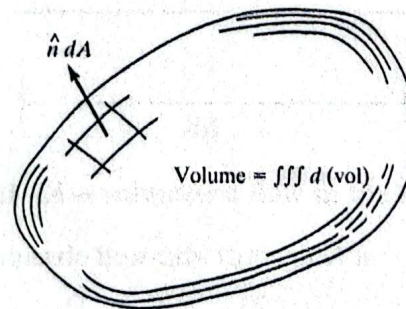
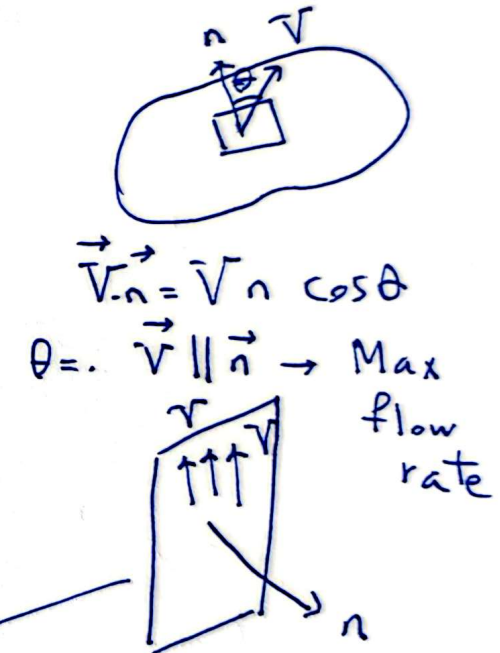


Figure 1.4: Fig. 2.5 from AE-BC

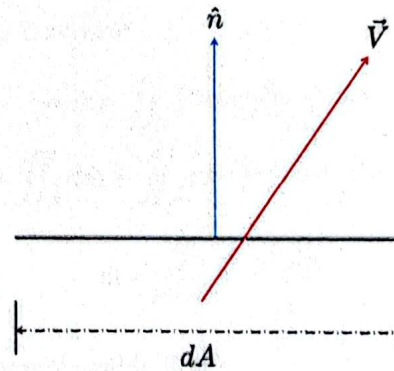
AE 319, EPG Cons. Eqns., Basics, 6



$$\theta = 90^\circ \rightarrow \vec{V} \perp \vec{n}$$

Mass flow is zero

### Volumetric flow rate



$$dQ\Delta t = \vec{V} \cdot \hat{n} \Delta t dA \rightarrow \text{volumetric flow in the } \hat{n} \text{ direction} \quad (1.1)$$

Therefore the volumetric flow rate through  $A$  is

$$Q = \iint_A dQ = \iint_A \underbrace{\vec{V} \cdot \hat{n}}_{\text{volume flux}} dA = \iint_A (\text{volume flux}) dA \quad (1.2)$$

$$\dot{m} = \iint_A (\text{mass flux}) dA$$

$$= \iint_A (\rho \times \text{volume flux}) dA$$

$$\dot{m} = \iint \rho \vec{V} \cdot \hat{n} dA = \rho \iint \vec{V} \cdot \hat{n} dA$$

$$= \rho Q$$

Mass flow rate

Similarly, the mass flow rate

$$\rightarrow \dot{m} = \iint_S (\text{mass flux}) dA$$

$$\dot{m} = \iint_A d\dot{m} = \iint_A \rho \times (\text{volume flux}) dA$$

$$\dot{m} = \iint_A \rho \vec{V} \cdot \hat{n} dA \quad (1.3)$$

if  $\rho = \text{constant}$ , (incompressible flow),

$$\dot{m} = \rho \iint_A \vec{V} \cdot \hat{n} dA = \rho Q \quad (1.4)$$

conservation of mass: rate of mass change in the volume is equal to the rate of mass transport through the surfaces.

### Conservation of Mass

Statement of conservation of mass: Mass is neither created nor destroyed

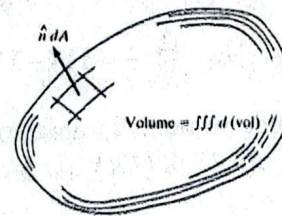
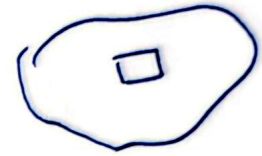


Figure 1.5: Fig. 2.5 from AE-BC

$$\underbrace{\{\text{rate of mass accumulation in volume}\}}_I = \underbrace{\{\text{rate of mass flow into volume}\}}_{II} - \underbrace{\{\text{rate of mass flow out of volume}\}}_{III}$$

mass <sup>rate</sup> change in the volume:  $\frac{\partial}{\partial t} (\dot{m})$ , ( $m = \rho \text{ Vol}$ )



$$= \frac{\partial}{\partial t} (\rho \text{ Vol})$$

$$= \frac{\partial}{\partial t} \left( \iiint_{\text{Vol}} \rho d(\text{Vol}) \right)$$

Conservation of mass (TSec. 2.2)

$$I \rightarrow \frac{\partial}{\partial t} (\text{mass}) = \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho d(\text{vol}) \quad (1.5)$$

$$II - III = - \iint_A \rho \vec{V} \cdot \hat{n} dA \quad (1.6)$$

Therefore, global conservation of mass or the integral form of the conservation of mass from eq. (1.5) & (1.6),

$$\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho d(\text{vol}) + \iint_A \rho \vec{V} \cdot \hat{n} dA = 0 \quad (\text{TEq. (2.5)}) \quad (1.7)$$

if the volume is fixed,

$$\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho d(\text{vol}) = \iiint_{\text{vol}} \frac{\partial \rho}{\partial t} d(\text{vol}) \quad (1.8)$$

mass transport through the surface is

$$\iint \rho \vec{V} \cdot \vec{n} dA$$

In general rate of mass change in the volume + mass transport through the surface is zero.

Integral form

$$\left( \frac{\partial}{\partial t} \iiint_{\text{Vol}} \rho d(\text{Vol}) + \iint_A \rho \vec{V} \cdot \vec{n} dA = 0 \right) \Rightarrow \iint \frac{\partial \rho}{\partial t} d(\text{Vol}) + \iint \rho \vec{V} \cdot \vec{n} dA = 0$$

## Conservation of mass

Divergence theorem:

$$\oint_A \vec{F} \cdot \hat{n} dA = \iiint_{\text{vol}} \nabla \cdot \vec{F} d(\text{vol}) \quad (1.9)$$

where,  $\vec{F}$  is any vector. Using eq. (1.9) in eq. (1.6) we get,

$$\oint_A (\rho \vec{V}) \cdot \hat{n} dA = \iiint_{\text{vol}} \nabla \cdot (\rho \vec{V}) d(\text{vol}) \quad (1.10)$$

Substituting eq. (1.10) in eq. (1.7) we get,

$$\iiint_{\text{vol}} \left[ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) \right] d(\text{vol}) = 0 \quad (1.11)$$

The volume is fixed, but it has been chosen arbitrarily hence it has to be true everywhere .

AE 319, EPG Cons. Eqns., Cont. Eq., 11

$$\begin{aligned} \iiint \frac{\partial \rho}{\partial t} d(\text{vol}) + \iint (\rho \vec{V}) \cdot \hat{n} dA &= \iiint \frac{\partial \rho}{\partial t} d(\text{vol}) + \iiint \nabla \cdot (\rho \vec{V}) d(\text{vol}) \\ &= \iiint \left( \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) \right) d(\text{vol}) = 0 \Rightarrow \left( \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) \right) = 0 \\ \iiint_{\text{Volume}} f(\vec{x}) &= 0 \Rightarrow f(x) = 0 \end{aligned}$$

Differential form

$$\nabla \cdot (\vec{F}) = \frac{\partial}{\partial x} (F_x) + \frac{\partial}{\partial y} (F_y) + \frac{\partial}{\partial z} F_z$$

### Conservation of mass

We can also shrink the volume to a point (every point in a flow field).  
For eq. (1.11) to be true then,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (\text{TEq. (2.2)}) \quad (1.12)$$

everywhere, which is known as the **local conservation of mass** or the **differential form of the conservation of mass**.

$$\nabla \cdot (\rho \vec{V}) = \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) \quad (\vec{V} = u\hat{i} + v\hat{j} + w\hat{k})$$

AE 319, EPG Cons. Eqns., Cont. Eq., 12

$$\text{Continuity Eq.} = \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} (\rho u) + \frac{\partial}{\partial y} (\rho v) + \frac{\partial}{\partial z} (\rho w) = 0$$

$$\text{Assume } \rho = \text{cte} \quad \Rightarrow \rho \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = 0 \Rightarrow \boxed{\nabla \cdot \vec{V} = 0}$$

"Incompressible flow"

For 2D flows,  $\vec{V} = u\hat{i} + v\hat{j}$

$$\Rightarrow \nabla \cdot \vec{V} = . \rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = .$$

Example: A 2D, incomp. flow has  $u = 4x$  component, find  $\vec{v}$ .

Solution:

$$\nabla \cdot \vec{V} = . \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = . \Rightarrow \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}, \quad u = 4x, \quad \frac{\partial u}{\partial x} = 4$$

$$\Rightarrow \frac{\partial v}{\partial y} = -4 \rightarrow v = -4y + C_1$$

Assume at  $y = 0$ ,  $v = 0 \rightarrow \boxed{C_1 = 0}$

$$\Rightarrow \begin{cases} u = 4x \\ v = -4y \end{cases}$$

$\vec{\nabla} \cdot \vec{V}$ : Gradient of  $v$   
 $\vec{\nabla} \cdot \vec{V}$ : Divergence of  $\vec{V}$   
 $\vec{\nabla} \times \vec{V}$ : curl of  $\vec{V}$

$$\boxed{\nabla \cdot \vec{V}}$$

$$\nabla \cdot \vec{V} = \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 4 - 4 = 0$$

well, it's incompressible

$$\vec{\nabla} \cdot \vec{V} : \text{Scalar} \rightarrow \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \vec{\nabla} \cdot \vec{V}$$

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} : \text{rotation, vorticity}$$

Divergence of  $\vec{V}$   
continuity Eq<sup>n</sup>

curl of  $\vec{V}$

$\vec{\nabla} \vec{V} \rightarrow$  is a tensor

Gradient of  $\vec{V}$

$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$m\vec{V}$  : Momentum

Force  $\equiv$  rate of change of momentum

~~Conservation of Momentum~~ (TSec. 2.3)

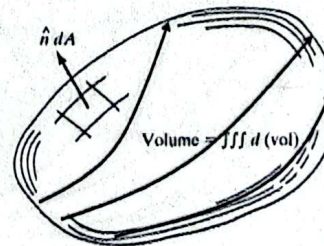


Figure 1.6: modified Fig. 2.5 from AE-BC

State of conservation of momentum: Newton's second law.  
Mathematically expressed as

$$\vec{F} = m\vec{a} = \frac{d}{dt}(m\vec{V}) \quad (1.14)$$

where  $\vec{F}$  is the force on the fluid,  $\vec{a}$  the acceleration and  $m$  the mass.

## Conservation of Momentum

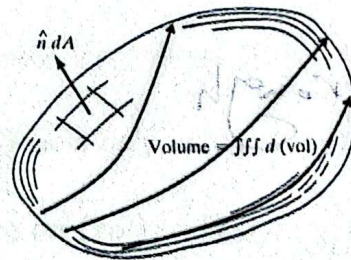


Figure 1.7: modified Fig. 2.5 from AE-BC

$$\begin{aligned}
 & \{\text{rate of momentum accumulation in (vol)}\} \\
 = & \{\text{rate of momentum in}\} - \{\text{rate of momentum out}\} + \\
 & \{\text{sum of forces acting on the system}\} \\
 = & -\{\text{net rate of momentum flow out of (vol)}\} \\
 & + \{\text{sum of forces acting on the system}\} \quad (1.15)
 \end{aligned}$$

## Cons. of Mass

rate of change of  
mass in volume (vol.)

+

mass transport through  
the surfaces

||

0

## Transport of Mom.

rate of change  
of Mom ( $m\bar{v}$ ) in Vol.

+

Mom. transport through  
the surfaces

||

Force